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SatQuery AI — ask your satellite in plain language

Problem Statement ID – 26167

Problem Statement Title – SatQuery AI: An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries

Theme – Space Technology

PS Category – Software

Team ID – [to be filled]

Team Name – x64

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SATQUERY AI

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Proposed solution (idea / solution / prototype) · how it addresses the problem · innovation & uniqueness

Proposed Solution / Approach

Agentic by design: a natural-language query is planned into a sequence of specialist tools drawn from a predefined registry (PS P6–P8) — not one monolithic VLM.

Guardrailed execution: schema + parameter whitelist and a deterministic fallback router; incompatible inputs return a compatibility report — never a crash.

Auditable trace (the graded surface): every run logs task, tools + versions, parameters, input hashes, latency and confidence (P9–P10).

Evidence, not vibes: answers ship with change masks, grounding boxes and caption overlays, plus calibrated High/Med/Low badges and an optical↔SAR disagreement flag.

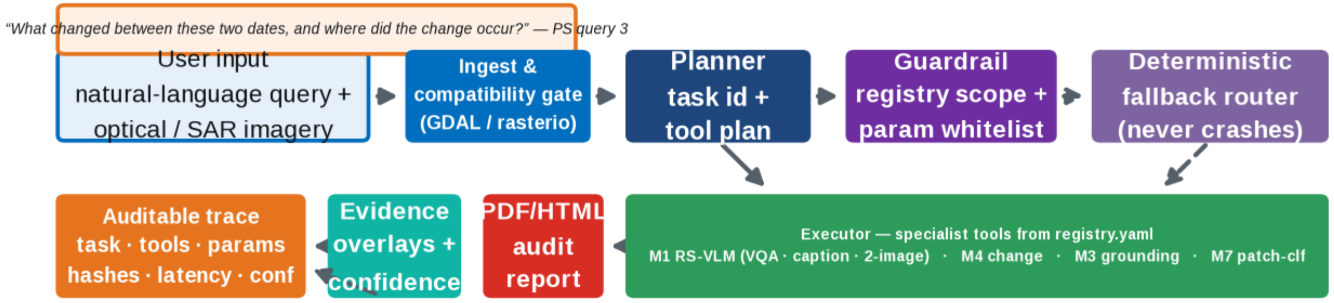
One-click audit report: downloadable PDF/HTML report per query — trace + evidence + confidence, ready for evaluation.

How it addresses the problem: *turns expert GIS pipelines into a query box, and makes every answer checkable — the trust gap the PS itself names.*

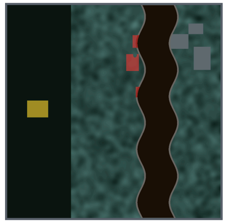
Representative PS queries we demo (verbatim):

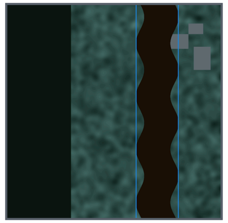
1 "Describe the land-cover and major objects visible in this image."
2 "Highlight the water body referred to in the query."
3 "What changed between these two dates, and where did the change occur?"
4 "Use the optical and SAR images together to identify built-up and water-covered regions."
5 "Has the built-up area increased, decreased, or remained unchanged?"

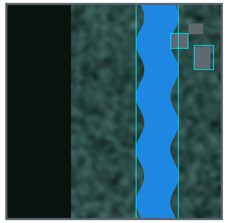
SATQUERY AI — end-to-end agentic workflow (every step logged to trace.json)



REAL PROTOTYPE OUTPUTS — thin end-to-end build (Window A)


change mask


grounding boxes


caption overlay

Innovation & Uniqueness

Trace as a product
the graded surface is a first-class UI + JSON artifact

Registry + guardrails
tools declared, params whitelisted, routing deterministic

Optical × SAR joint reasoning
optical gives spectra & context; SAR sees through cloud, day & night — paired focus (P1, P5)

RS-adapted, never generic
QLoRA on BigEarthNet.txt — generic VLMs explicitly insufficient (P12)

Honest confidence
calibrated badges, disagreement flags, reasoned abstention

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TECHNICAL APPROACH

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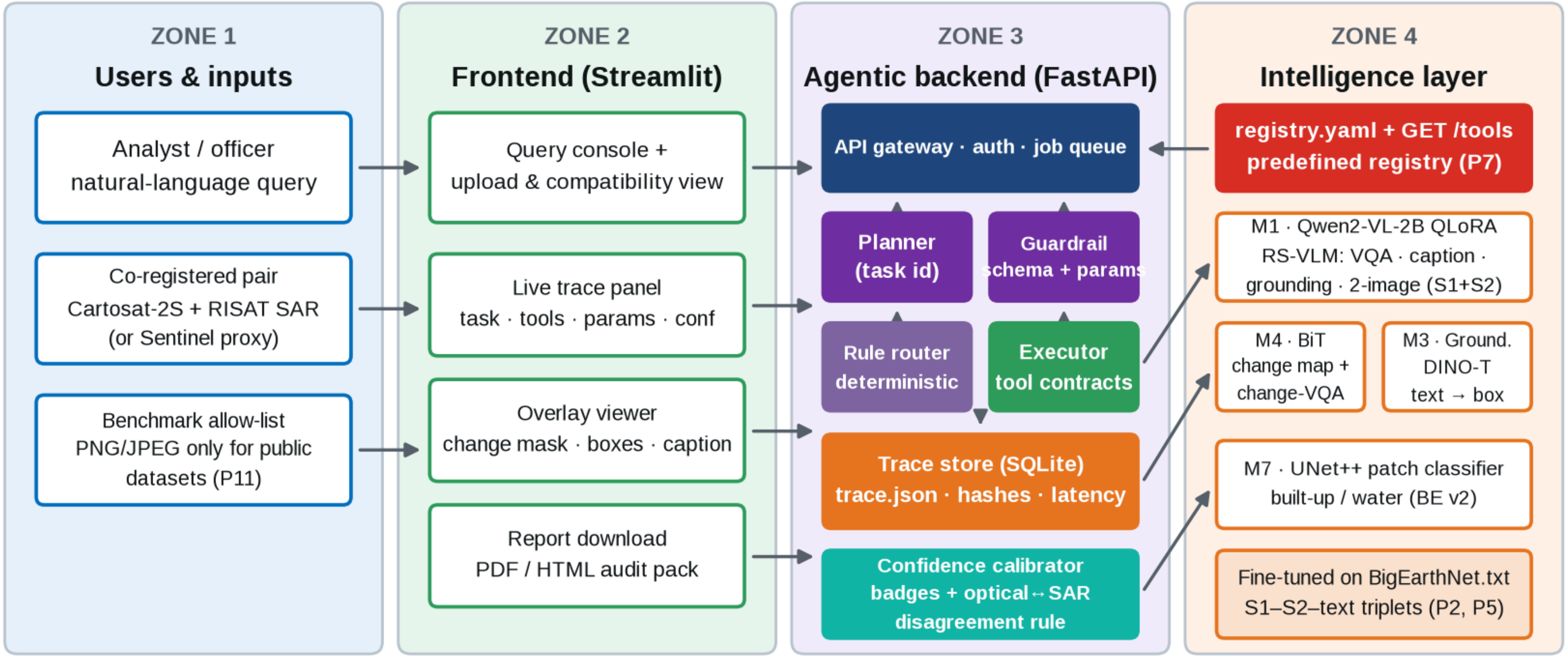
Technologies to be used · methodology & process for implementation (flow charts / images / working prototype)

System architecture — sensor-agnostic 512 px tiles / 64 px overlap, end-to-end



Implementation Process

1. **Ingest & validate**
   format · CRS · modality · co-registration · 512/64 tiling (GDAL)
2. **Plan & guard**
   task id, tool selection from registry, parameter whitelist
3. **Execute**
   specialists run per tile; every call writes to the trace
4. **Fuse & calibrate**
   weighted geometric mean, High/Med/Low badges, disagreement rule
5. **Visualise**
   change mask / boxes / caption streamed over the scene
6. **Report**
   PDF/HTML audit pack + trace.json download

Technology stack to be used

Python · FastAPI · Streamlit · GDAL / rasterio · PyTorch · Qwen2-VL (QLoRA) · Grounding DINO · UNet++ / BiT · Docker

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FEASIBILITY AND VIABILITY

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Feasibility analysis · potential challenges & risks · strategies for overcoming these challenges

Feasibility

- **Proven open weights & open source —** Rs 0 licence cost.
- **Prototype already runs end-to-end** on a laptop (thin build, Window A).
- **QLoRA 4-bit + 2B backbone** fits free Kaggle / Colab GPUs.
- **Tile-based 512 px / 64 overlap** → sensor- & resolution-agnostic.

Viability

- **15-day dated build plan** with daily checkpoints D1–D18.
- **3-model triage (M1/M4/M3)** matches pooled free compute exactly.
- **Demo-day safe:** deterministic fallback + cached-answer mode.
- **Metrics gates** on public splits before every submission.

Practical Implementation

- **Dockerised, offline deploy** for the evaluation venue.
- **Handles georeferenced Cartosat-2S + RISAT** pairs by design (P14).
- **Normalised metric tables** (P15) in the final report.
- **Same tool contracts** from prototype → trained models.

Potential Challenges & Risks	Strategies For Overcoming Challenges
01 **Domain gap:** trained on Sentinel (10–20 m), evaluated on Cartosat-2S (~1 m) + RISAT.	**Tile-based, resolution-agnostic pipeline;** week-1 proxy tests on public VHR + L-band. 01
02 **Tight 6 GB VRAM** at the evaluation venue.	**QLoRA 4-bit + 2B backbone;** heavy training on free 16 GB Kaggle GPUs. 02
03 **Hidden eval annotations** & unpublished metric weights (G1).	**Train on the PS-named public benchmarks;** report normalised metric tables (P15). 03
04 **Demo-day failure risk** in front of judges.	**Guardrail fallback router + cached answers;** honest, scoped narration on stage. 04
05 **15 days for 3 trained models.**	**Binding triage M1/M4/M3;** dated checkpoints D1–D18; contracts frozen early. 05

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IMPACT AND BENEFITS

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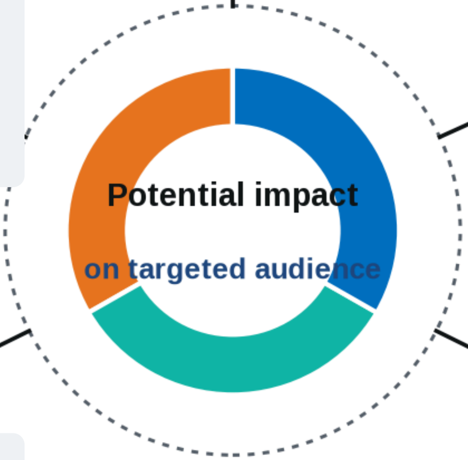
Potential impact on the target audience · benefits of the solution (social, economic, environmental)

ISRO / SAC analysts: task answers with evidence in minutes, not scripting days.

Disaster cells: bi-temporal change maps + change-VQA for rapid damage triage.

Agriculture dept: water / crop queries in plain language — no GIS staff needed.



Urban planners: auditable built-up growth counts from masks, not eyeballing.

Students & researchers: open traces + reports make RS science reproducible.

Benefits of the solution

Social	“Democratizes satellite intelligence — any officer queries in plain language, the PS's own stated goal.”
Economic	“One assistant replaces many single-task vendor pipelines; saves analyst-hours on every tasking cycle.”
Environmental	“Faster spotting of floods, encroachment and water-body change → quicker response on the ground.”

“Every answer with evidence, confidence and an audit trail — satellite intelligence anyone can query.”

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RESEARCH AND REFERENCES

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Details / links of the reference and research work

Official problem statement & portals

- **SIH 2026 Problem Statement 26167 (ISRO):** “SatQuery AI — An Interactive Vision-Language Assistant for Multimodal Remote Sensing Image Analysis through Text Queries” · sih.gov.in
- **ISRO / Space Applications Centre:** evaluation set of pre-georeferenced Cartosat-2S + RISAT pairs (P14) · isro.gov.in

Datasets & benchmarks named by the PS

- **BigEarthNet.txt** (primary adaptation data — 9.55 M annotations over 464 K S1–S2 pairs): arXiv:2603.29630 · huggingface.co/datasets/BIFOLD-BigEarthNetv2-0/BigEarthNet.txt
- **VRSBench** (NeurIPS'24 D&B — captioning, grounding, VQA): arXiv:2406.12384 · github.com/lx709/VRSBench
- **RSVQA** (Lobry et al., IEEE TGRS 2020) — single-image VQA volume.
- **CDVQA** (Yuan et al., TGRS 2022 — bi-temporal change VQA + semantic change maps): arXiv:2112.06343 · github.com/YZHJessica/CDVQA
- **TAMMI (2025)** — VQA over co-located VHR RGB + multispectral + SAR triplets; LEVIR-CD / WHU-CD for change-mask volume (github.com/justchenhao/Levir-CD).

Models & methods

- **Qwen2-VL** (2 B backbone, QLoRA 4-bit adaptation): arXiv:2409.12191 · QLoRA arXiv:2305.14314 · LoRA arXiv:2106.09685
- **Grounding DINO** (text-guided grounding base): arXiv:2303.05499
- **UNet++ / BiT** (change & patch-classifier heads): arXiv:1807.10165

Our groundwork (this submission)

- **Clause-level PS analysis:** all 15 binding clauses (P1–P15) mapped to design decisions; 5 known PS gaps logged (G1–G5).
- **Working thin prototype:** real ingestion (GDAL), registry + guardrail + fallback router, trace store, overlays and PDF reports — screenshots on slide 2.
- **Dated 15-day build plan:** three trained models (M1/M4/M3) with daily checkpoints and benchmark gates for the 20 Sep final submission.

Benchmark gates for the 20 Sep submission — zero-shot → adapted delta is our P12 compliance proof

Captioning	Grounding	Single-image VQA	Change-VQA	Change map
VRSBench test	VRSBench refs	RSVQA + VRSBench	CDVQA test	CDVQA / QAG-360K
CIDEr · BLEU-4 · METEOR · ROUGE-L	**box IoU ≥ 0.5 acc · COCO AP**	**exact match · soft accuracy**	**exact match over answer categories**	**IoU · F1 · mIoU**

All scores normalised before combining (P15); per-tool latency and confidence recorded in every trace.